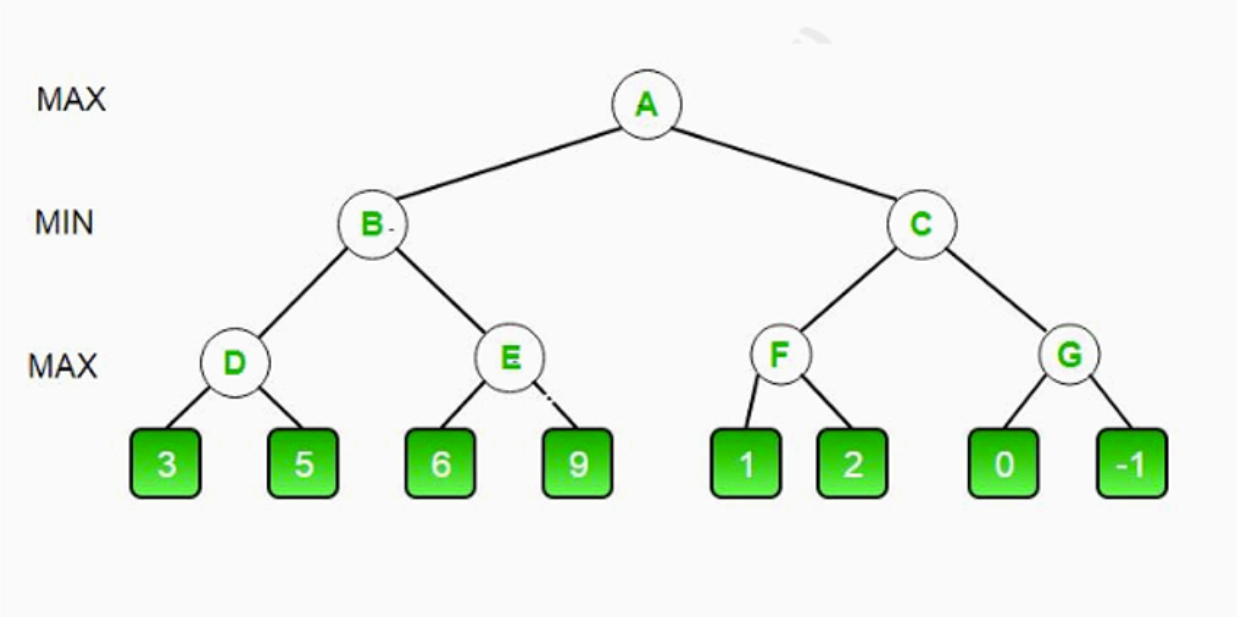
# Que : 1




Step 1: Start at A (MAX)

- A wants the **largest value**
- It compares **B and C**
- It checks **B first**

Step 2: Go to B (MIN)

- B wants the **smallest value**
- It compares **D and E**
- It checks **D first**

Step 3: Go to D (MAX)

- D sees two values: **3 and 5**
- MAX chooses the bigger one → **5**
- D returns **5** to B

Step 4: Back at B

- B now knows it can get **5**
- B checks **E** to see if it can get **less than 5**
-

Step 5: Go to E (MAX)

- First value seen = **6**
- 6 is **already bigger than 5**

**STOP here!**

Why?

- MIN at B already has **5**
- MAX at E gives **6**
- MIN will **never choose 6**
- So the next value (**9**) is useless

**E returns 6**

Step 6: Final value at B

- MIN chooses the smaller of **5 and 6**
- So **B = 5**

Step 7: Back at A (MAX)

- A now has **5**
- A checks **C** to see if it can get something **better**

Step 8: Go to C (MIN)

- C checks **F first**

Step 9: Go to F (MAX)

- Values are **1 and 2**
- MAX chooses **2**
- F returns **2** to C

Step 10: Back at C

- C already has **2**
- But A already has **5**

**STOP here!**

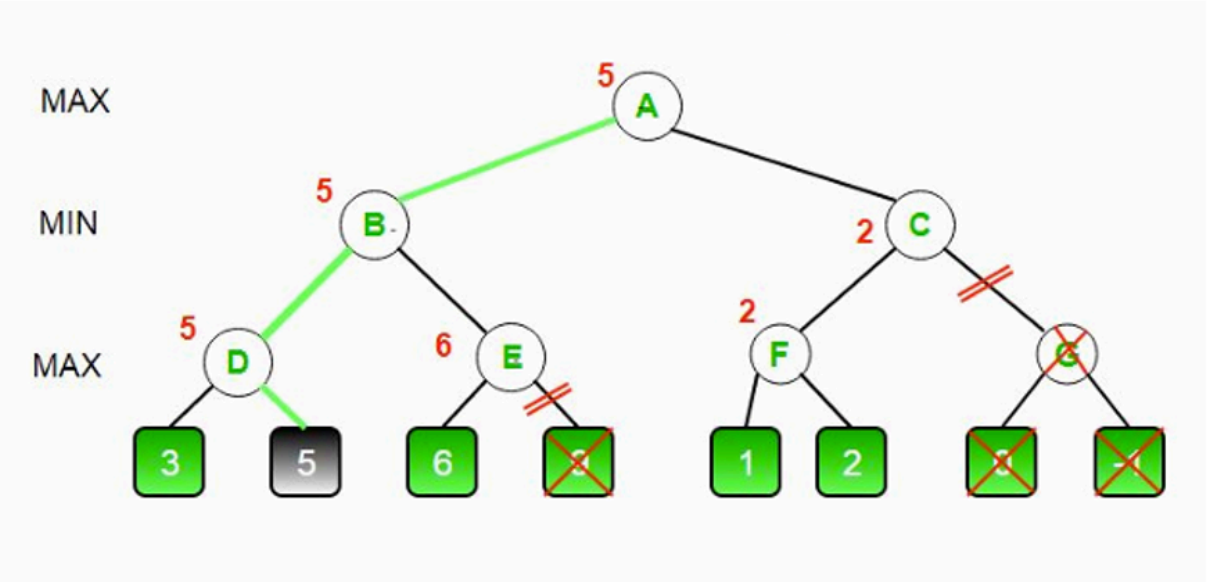
Why?

- MAX will never choose **2** when **5** is available
  So C does **not** check G

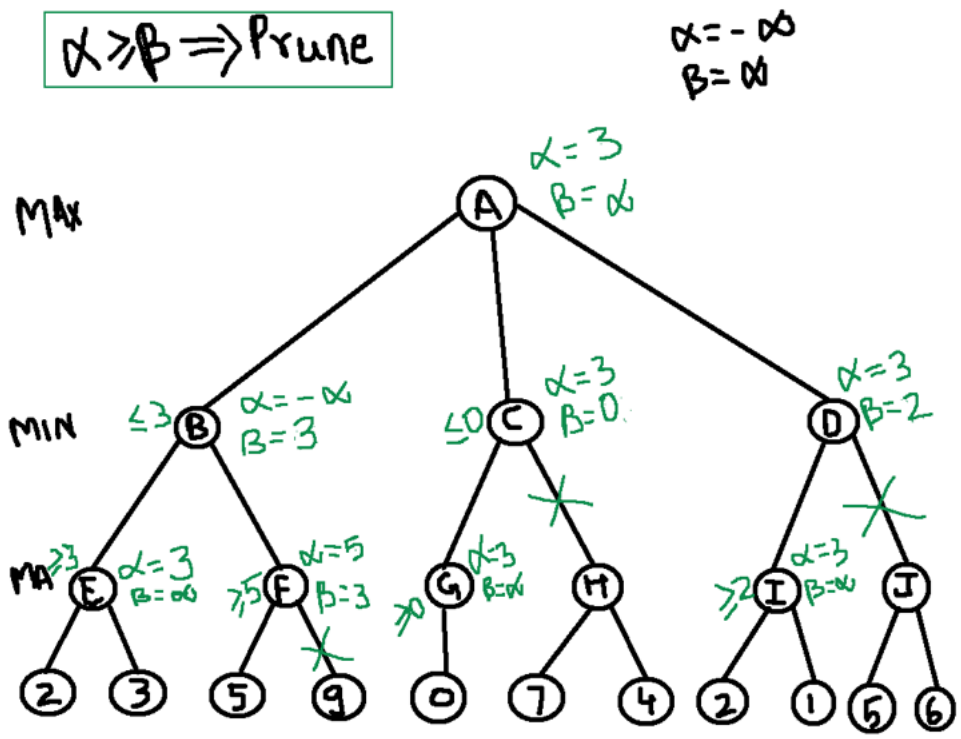
**C returns 2**

Step 11: Final Answer at A

- A compares **5 and 2**
- MAX chooses **5**



# Que : 2




Root **A = MAX**
Level 2 = **MIN** (B, C, D)
Level 3 = **MAX** (E, F, G, H, I, J)

Rule used: **If α ≥ β ⇒ Prune**

Initial values:
$\alpha = -\infty$, $\beta = +\infty$

Step 1: Node A (MAX)

A selects the maximum value from its children.

Step 2: Node B (MIN)

Step 3: Node E (MAX) under B

Leaf values: 2, 3
MAX selects 3
E returns 3

At B:
$\beta = \min(+\infty, 3) = 3$

Step 4: Node F (MAX) under B

Leaf values: 5, 9
MAX sees 5 → $\alpha = 5$

Pruning check:
$\alpha\ (5) \geq \beta\ (3) \rightarrow$ TRUE
Node 9 is pruned

F returns 5

At B:
$\beta = \min(3, 5) = 3$B returns 3 to A

Step 5: Back at A (MAX)

$\alpha = \max(-\infty, 3) = 3$

Step 6: Node C (MIN)

Step 7: Node G (MAX) under C

Leaf value: 0
G returns 0

At C:
$\beta = \min(+\infty, 0) = 0$

Pruning check:
$\alpha\ (3) \geq \beta\ (0) \rightarrow$ TRUE
Entire H subtree (7, 4) is pruned

C returns 0 to A

Step 8: Back at A (MAX)

$\alpha = \max(3, 0) = 3$

Step 9: Node D (MIN)

Step 10: Node I (MAX) under D

Leaf values: 2, 1
MAX selects 2
I returns 2

At D:
$\beta = \min(+\infty, 2) = 2$

Pruning check:
$\alpha\ (3) \geq \beta\ (2) \rightarrow$ TRUE
J subtree (5, 6) is pruned

D returns 2 to A

Step 11: Final Decision at A (MAX)

Values received by A:
From B = 3
From C = 0
From D = 2

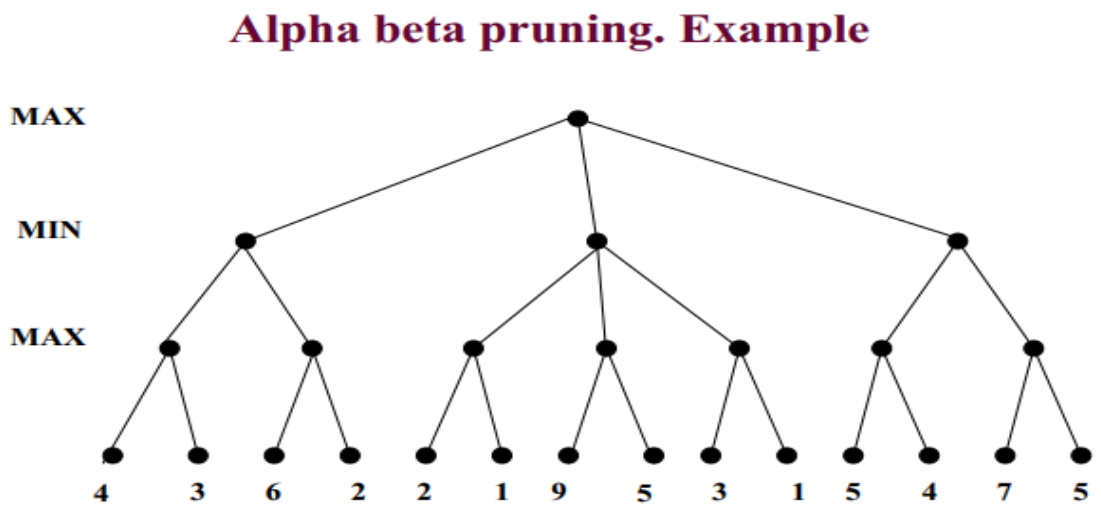
Final value:
max(3, 0, 2) = **3**

Final Answer

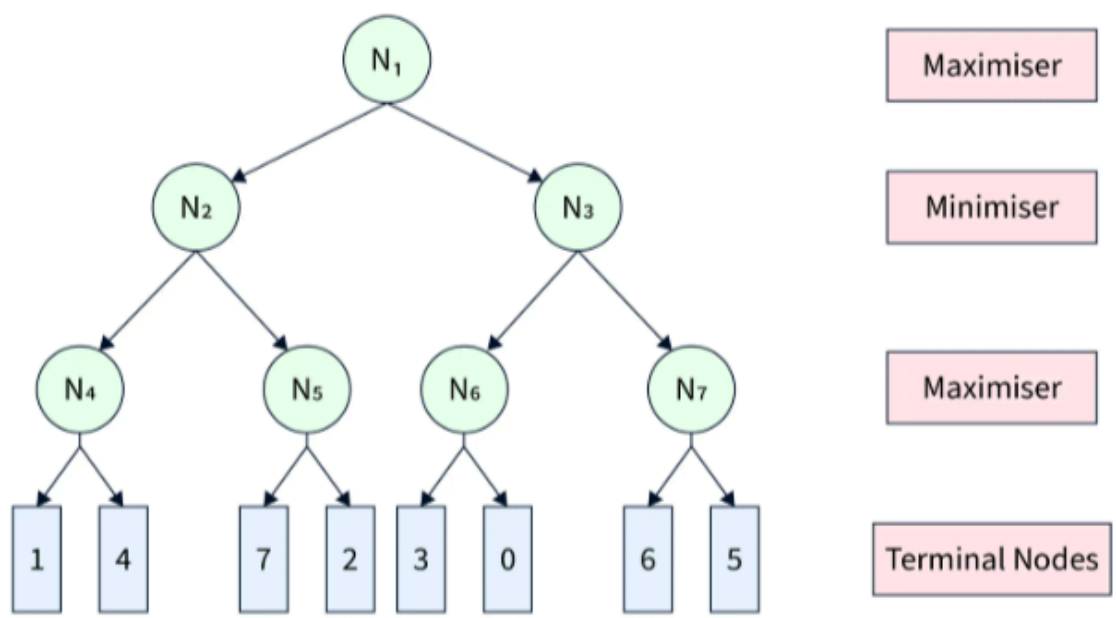
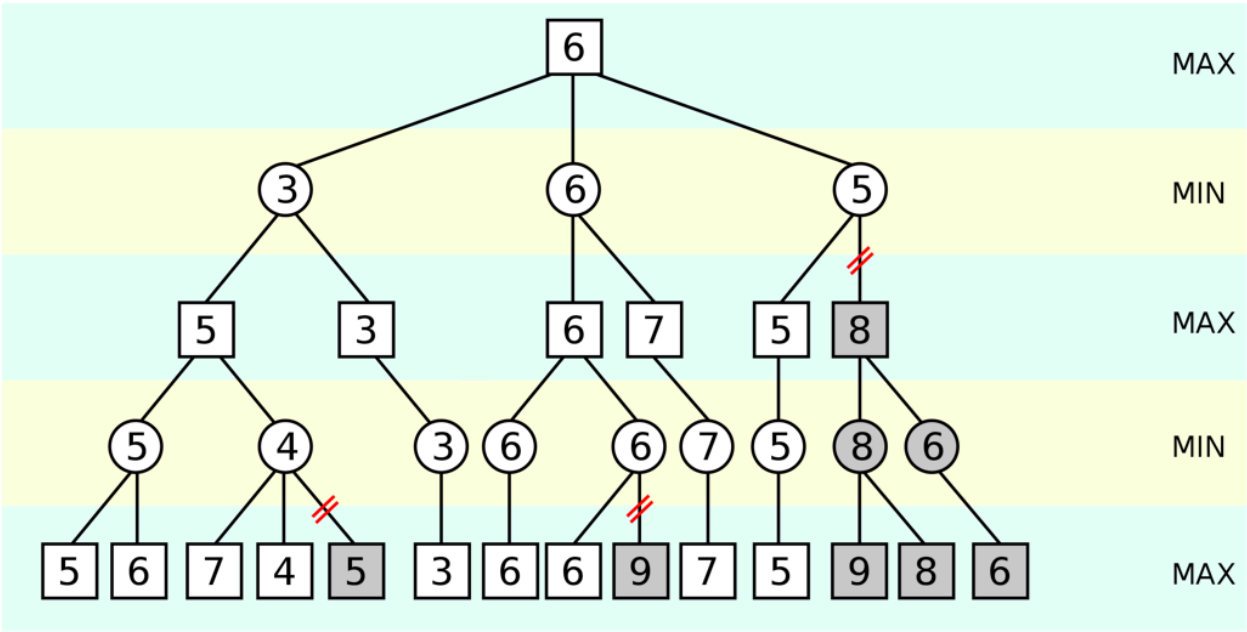
Optimal value at root **A = 3**

Pruned nodes:
• 9 (under F)
• H subtree (7, 4)
• J subtree (5, 6)

Example 1




Example 2



Example 3